



**Institute of Arts and Sciences (Chiniot Campus)
Govt College University Faisalabad**

Software Requirement Specification

(SRS Document)

For

Timetable Automation System

By

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Bachelor in Computer Science (2022-2026)

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of this document is to provide a detailed description of the Timetable Automation System, outlining its features, functionalities, and requirements. It serves as a guide for developers, testers, and other stakeholders involved in the development and maintenance of the system.

1.2 Intended Audience

This document is intended for:

- Developers
- Testers
- Project Managers
- Stakeholders involved in the Timetable Automation System

Readers are advised to review the document thoroughly to understand the requirements and constraints of the system.

1.3 Project Scope

The scope of this project is to provide a comprehensive solution for automating and optimizing the timetable management process, meeting the diverse needs of educational or organizational institutions.

1.4 Definitions

Student

In the context of the Timetable Automation System, a student refers to an individual who interacts with the system primarily to view the timetable. Students have the capability to log in, and view the generated timetable.

Instructor

An instructor is a user role responsible for creating and managing the Timetable Automation System. Instructors have administrative privileges that allow them to generate timetable and perform other administrative functions.

User

The term "user" refers to any individual, either a student or an instructor, who interacts with the Timetable Automation System. Users engage with the system through a user interface to perform their specific actions.

Administrator/Admin

An administrator, often referred to as admin, is a user with elevated privileges responsible for system configuration, user management, and overall system maintenance in the Timetable Automation System. Admins have access to features and settings beyond those available to regular users, such as creating and managing schedules, overseeing system operations, and resolving conflicts.

Input

Input is the data or information that a system receives from the external environment. It serves as the raw material for processing within a system.

In a computer program, user keyboard entries, mouse clicks, or data from sensors can be considered as input.

Output

Output is the result or information produced by a system or process after the input has been processed.

In a computer program, the output could be displayed on a screen, printed on paper, or saved in a file.

Prompt

Prompt refers to a message or indicator that prompts the user to provide input. It's a way for the program to communicate with the user.

Schedule

A schedule is a structured plan of events, tasks, or activities, arranged chronologically. In the Timetable Automation System, schedules are created to organize classes, allocate resources, and manage time efficiently for instructors and students.

Conflict

A conflict in the Timetable Automation System occurs when two or more events or tasks are scheduled at the same time and place, or when a resource is double-booked. The system identifies and helps resolve these conflicts to ensure smooth operations.

Allocation

Allocation refers to the process of assigning resources, such as classrooms, instructors, and equipment, to specific timeslots in the Timetable Automation System. Efficient allocation ensures optimal use of resources and smooth conduct of scheduled activities.

Availability

Availability indicates the times and dates when resources, such as instructors or classrooms, are free and can be scheduled for classes or other activities. The Timetable Automation System uses availability data to create feasible schedules.

Optimization

Optimization in the Timetable Automation System refers to the process of adjusting schedules to achieve the best possible arrangement of classes and resources, minimizing conflicts and maximizing resource utilization.

Timetable

A timetable is a detailed plan showing the times at which particular events, such as classes or meetings, will occur. In the Timetable Automation System, timetables are generated automatically based on input data and optimization algorithms.

2. Overall Description

2.1 Product Perspective

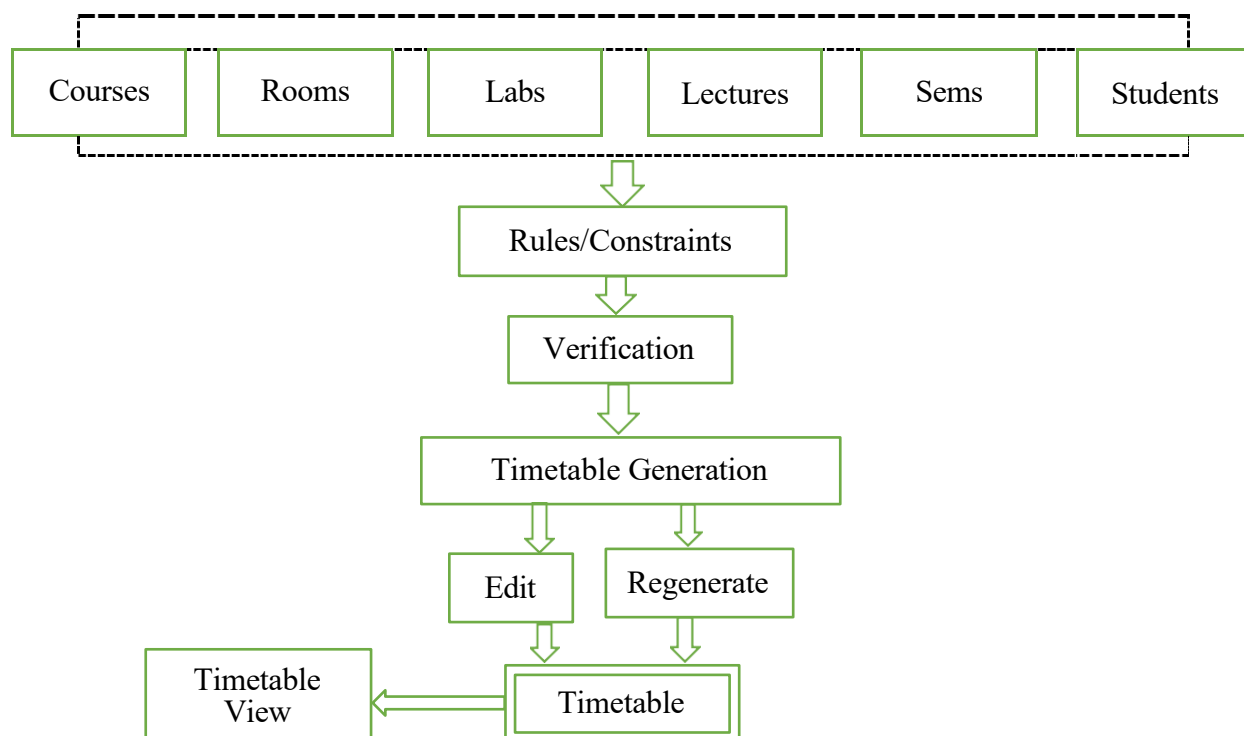
The Timetable Automation System will work as a standalone application with a console-based user interface. It will support different user roles, including students, and administrators to help create and manage timetables efficiently.

2.2 Product Features

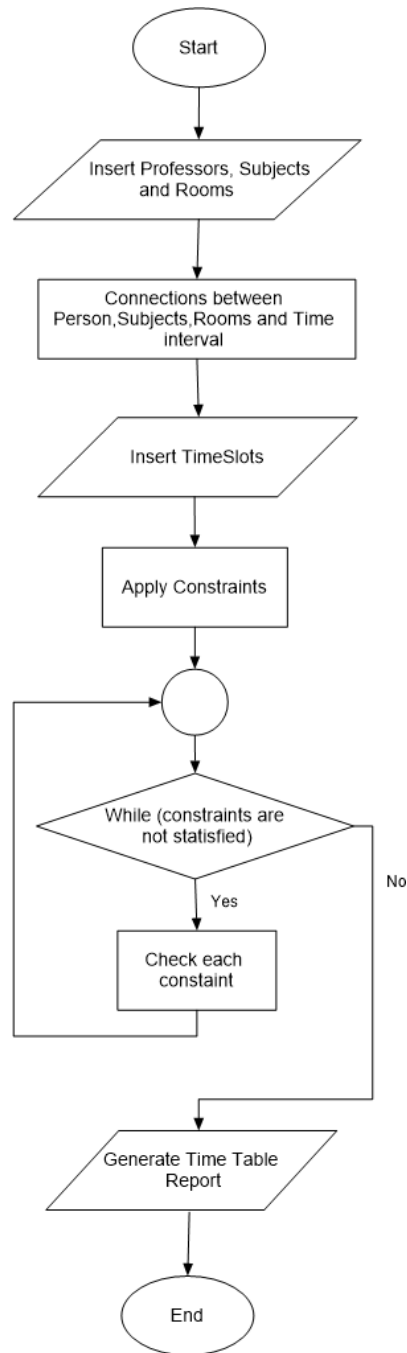
- User Authentication and Authorization
- Timetable creation and management
- Automatic generation of Timetable.
- Well formatted report in excel format.
- Resource Allocation
- User Role Management

2.3 User Classes and Characteristics

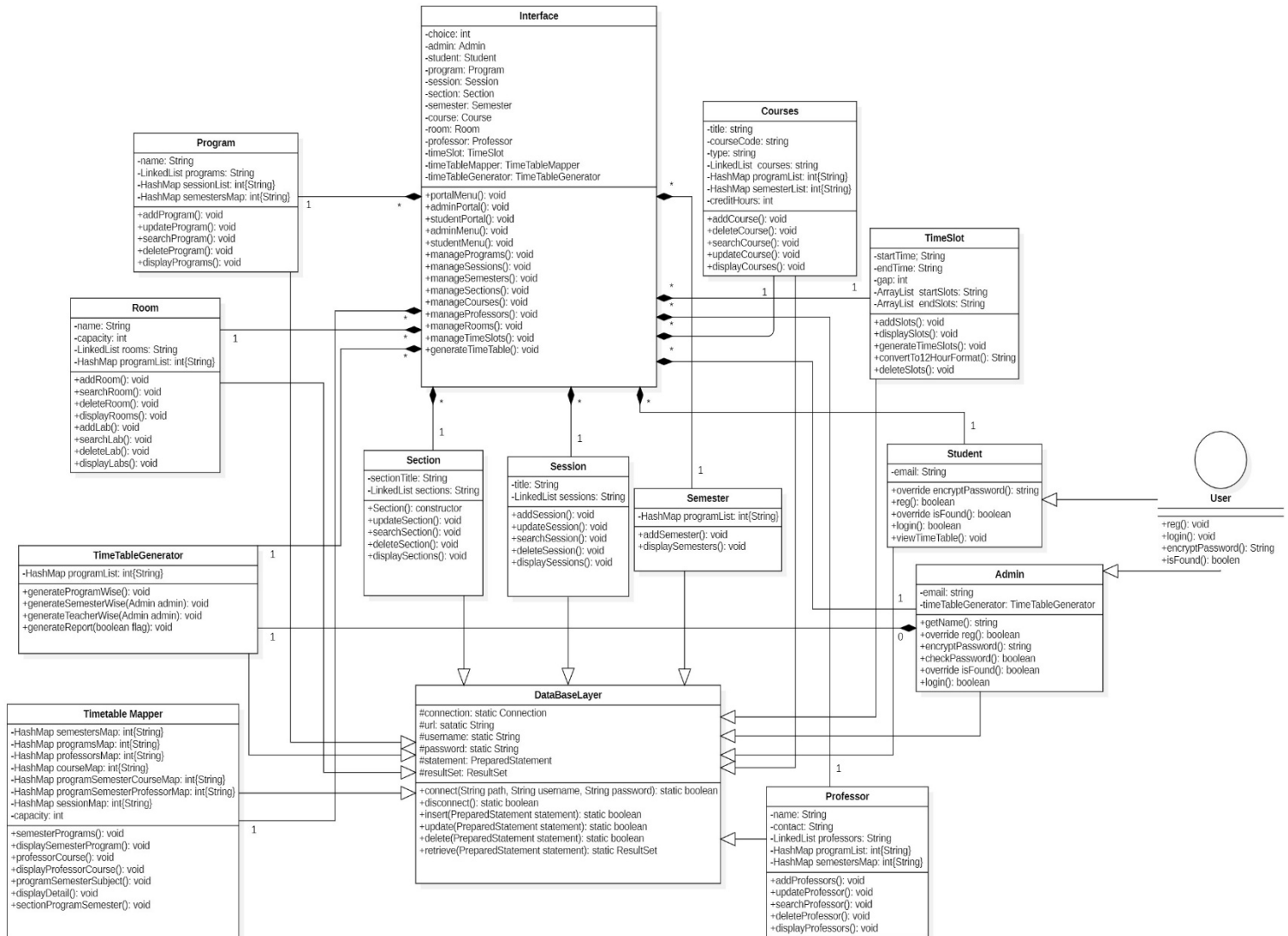
General View of Time Table Automation System



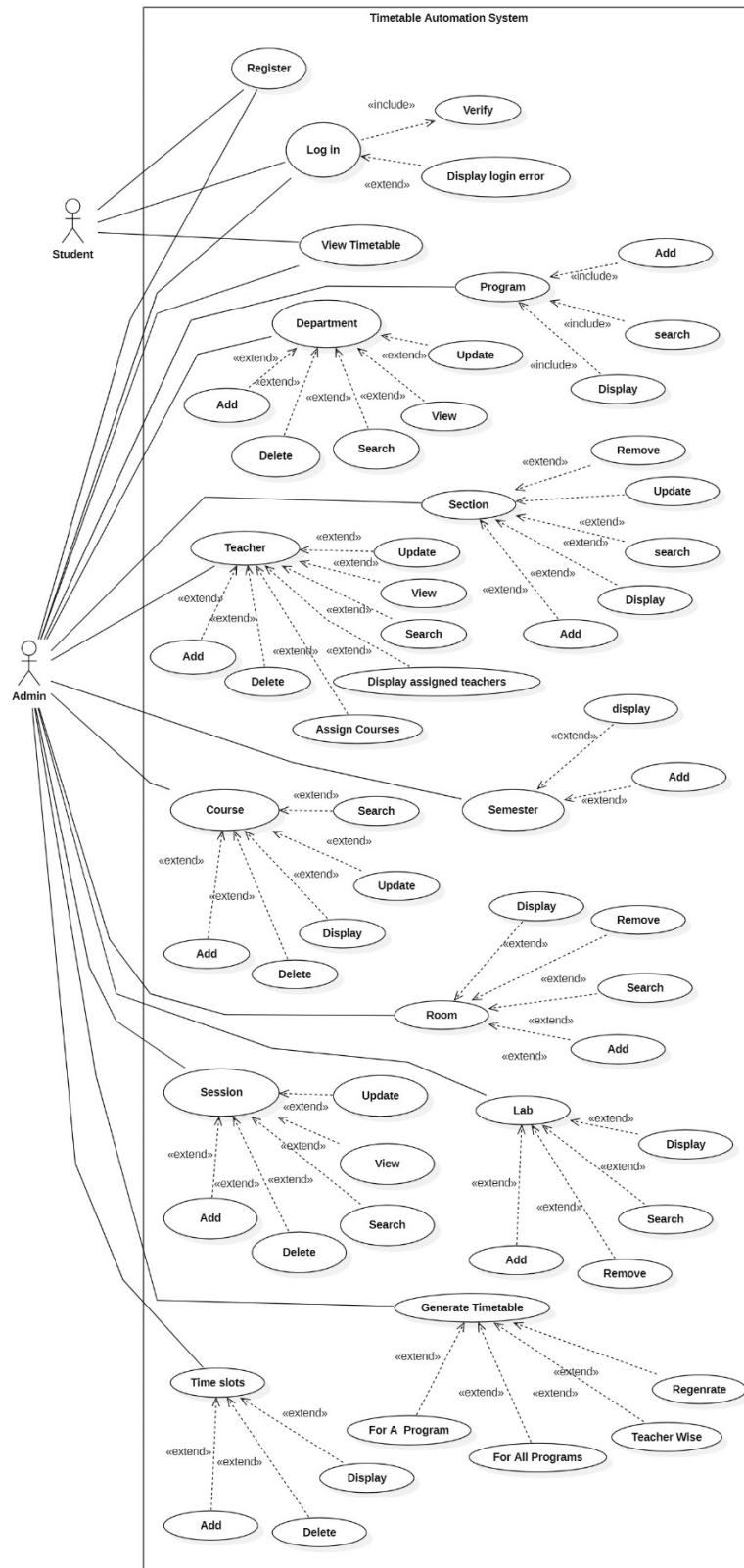
Algorithm



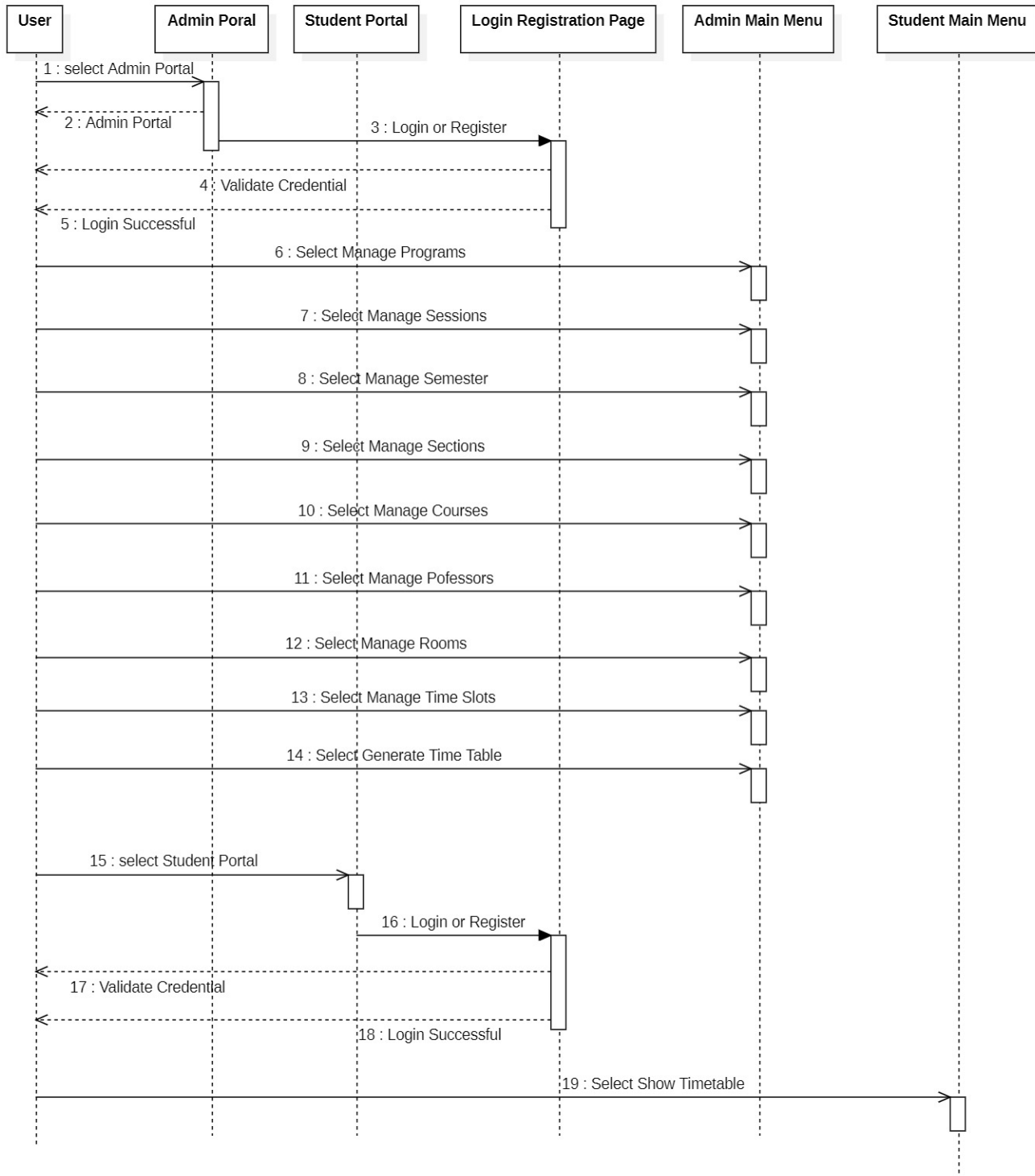
UML Diagram



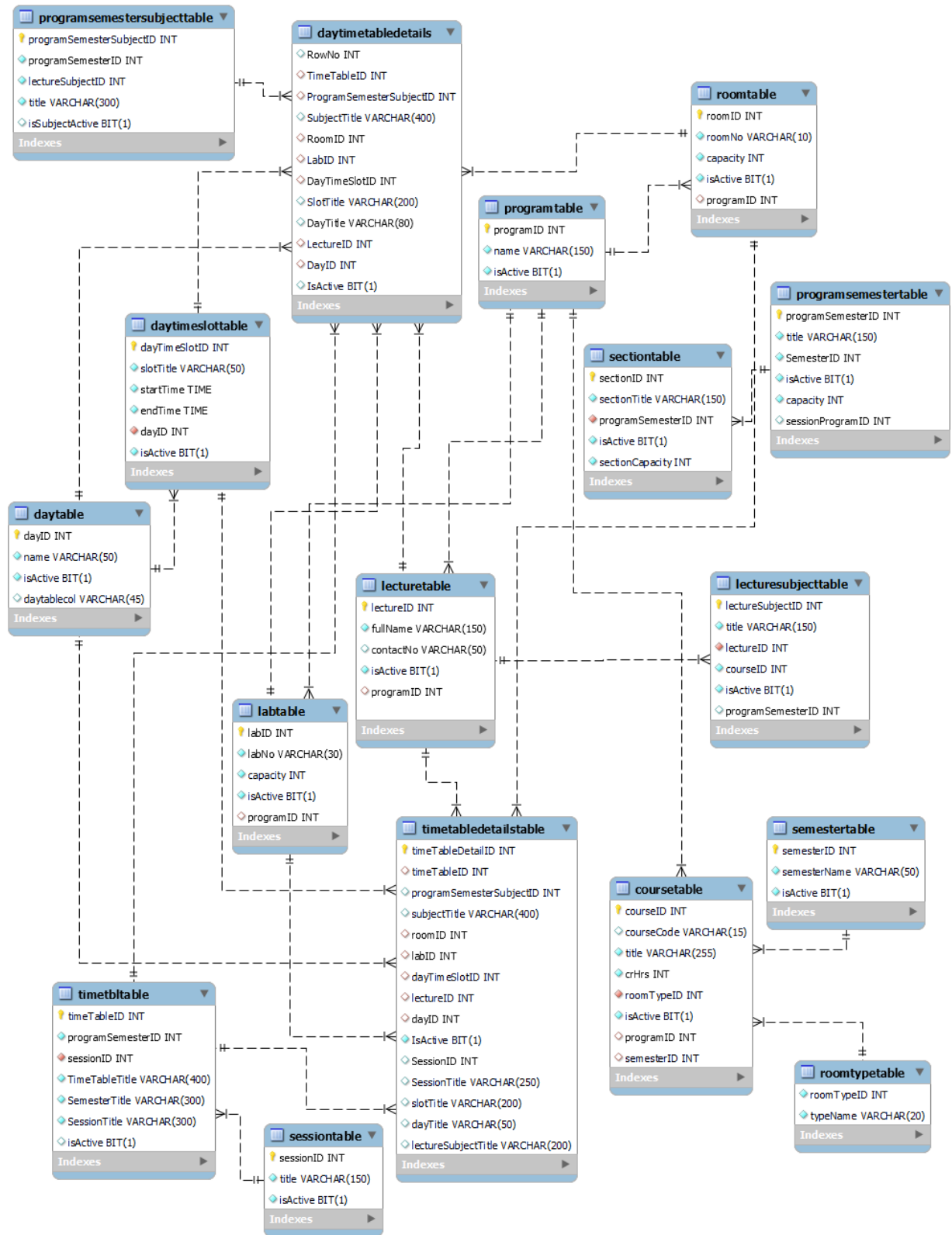
Use Case Diagram



System Sequence Diagram



ER Diagram



2.4 Operating Environment

The system will be compatible with modern web browsers (e.g., Chrome, Firefox, Edge) and will run on a server with Java support. The frontend will be accessible from any device with a web browser.

2.5 Design and Implementation Constraints

- The system will be developed using java and SQL Programming Language.
- SQL is used in the database.
- Compatibility with latest version of java and SQL.

2.6 User Documentation

Comprehensive user documentation will be provided, including user manuals and guides for both students and instructors.

2.6.1 Concepts

Object Oriented Programming (OOP) Principles:

- **Class**
 - Classes Used:
 - **User:** acts as an interface class for Student and Admin.
 - **Student:** Represents a student with a name.
 - **Admin:** Represents an admin with a email, name.
 - **Professor:** Use to store the data of professor.
 - **Room:** Manages information about rooms and labs.
 - **Subject:** Manages information about semester subjects.
 - **Semester:** Manages information about semesters in a program.
 - **Session:** Manages information about sessions.
 - **Timeslot:** Use to manage the timeslots.
 - **TimeTableGenerator:** Use to generate the timetable.
 - **TimeTableMapper:** Use to control the time mapping functionalities.
 - **Interface:** Provides a user interface for interacting with the Timetable Automation System.
- **Encapsulation**
 - **Private Members:** Various data members in classes are marked as private, encapsulating the internal details of each class.
 - **Explanation:** Encapsulation is the bundling of data and the methods (functions) that operate on the data into a single unit, i.e., a class. This helps in controlling access to the data by providing access specifiers (public, private, protected).
- **Abstraction**
 - Each class encapsulates specific functionality, exposing only essential details to the other parts of the program. We are using user class as abstract class for Student and Admin.
 - **Explanation:** Abstraction is the concept of simplifying complex systems by modeling classes based on the essential properties and behaviors.

- **Inheritance**

- Admin and student inheriting from User which is Interface class.
- All Classes inherits from Database Layer, indicating an "is-a" relationship.
- **Explanation:** Inheritance is the mechanism by which one class can inherit properties and behavior from another class promoting code reusability.

- **Composition**

- Interface class contains and uses objects of other classes, such as Admin and Room, Admin is composed of timetable generator class object as well.
- **Explanation:** Composition refers to the design principle where a class is composed of or contains objects of other classes, enabling the creation of more complex and reusable structures.

Data Structure & Algorithms (DSA) Principles:

- **HashMaps:** Utilized for mapping keys (IDs) to values (names or titles), HashMaps provide efficient retrieval of data based on a unique identifier.
- **LinkedList:** LinkedLists are employed to store a dynamic list of elements where each element is linked to the next one. They offer efficient insertion and deletion operations at any position in the list.
- **ArrayList:** ArrayLists are dynamic arrays that can resize themselves as elements are added or removed. They provide fast random access to elements and are suitable for scenarios where frequent access and modification of elements are required.
- **Brute Force Approach**
 - The brute force approach involves systematically trying every possible solution until the correct one is found. It's often used in scenarios where the problem space is small enough to explore all possible combinations.
 - **Explanation:** In the context of the Time Table Automation System, a brute force approach is applied to tasks such as timetable generation or resource allocation. when generating a timetable, the brute force approach involves exhaustively trying every possible combination of subjects, professors, and time slots until a feasible timetable is found.

Software Engineering Principles:

- **Development Approach**

- The Timetable Automation System follows an iterative development model, where the software development process is divided into small, manageable iterations. Each iteration results in a potentially shippable product increment, allowing for flexibility, adaptability, and continuous refinement based on stakeholder feedback.

- **DRY (Don't Repeat Yourself)**

- Avoiding duplicating code by encapsulating common functionalities within classes and methods.

- **Modularity**
 - **Separation of Concerns:** Each class represents a specific concern, promoting modularity. TimeTableMapper focuses on mapping functionality, Student on student details, Admin on admin details, and Interface on user interaction.
- **Code Reusability**
 - **Inheritance for Reusability:** Inheritance is used to reuse the functionality.
- **Robust Password Encryption:**
 - **Enhanced Security Measures:** The admin class and student class incorporates robust password encryption techniques to safeguard sensitive user credentials.
 - **SHA-256 Hashing:** User passwords are securely hashed using the SHA-256 hashing algorithm, rendering them irreversible and highly secure.
- **Maintainability**
 - **Structured Code:** The code is organized into classes, functions, and headers, enhancing readability and maintainability.
- **Readability**
 - **Descriptive Identifiers:** Variable and function names are chosen descriptively, contributing to code readability.
- **Documentation**
 - **Usage:** Comments, descriptive variable names.
 - **Explanation:** The code includes comments that provide information about the purpose of the code blocks. Descriptive variable and function names also contribute to self-documenting code.

2.7 Assumptions and Dependencies

- Users are assumed to have access to a computer with a Windows operating system.
- The system relies on standard input and output for user interactions.
- The console-based application assumes that users are familiar with basic command-line interactions.
- Dependencies include the availability of the Java Runtime Environment (JRE) and a database management system (DBMS) SQL on users' machines.
- Any changes in the Windows environment, command-line interface, Java version, or DBMS may affect the application's functionality.

3. System Features

3.1 Admin Portal

3.1.1 Description and Priority

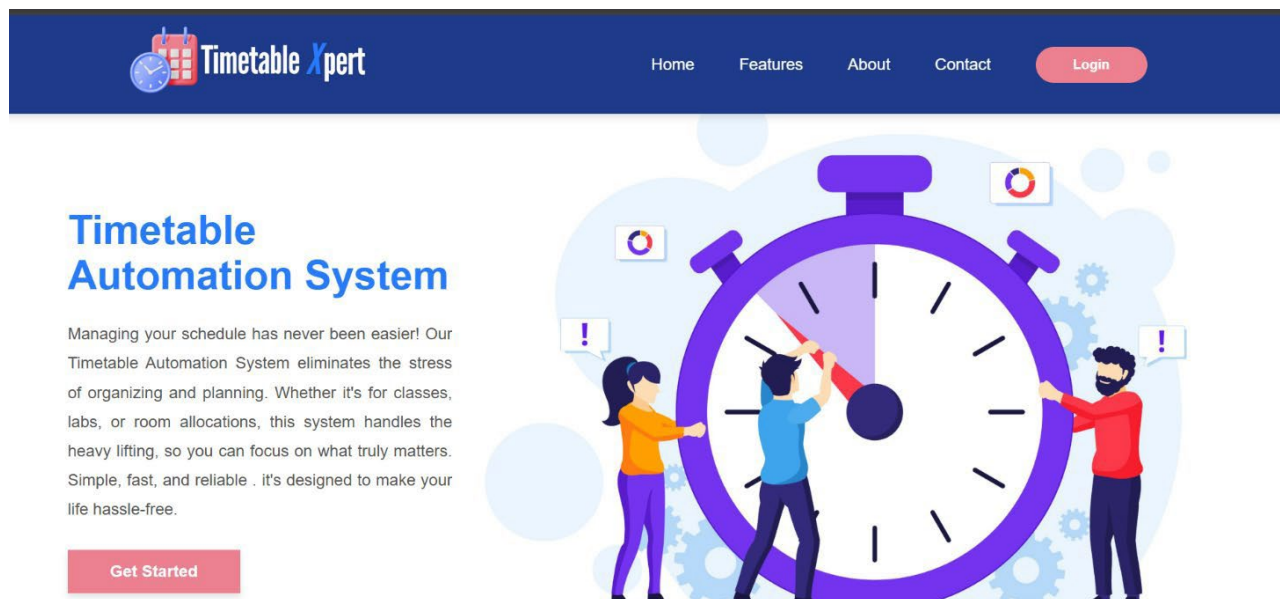
The Admin Portal enables administrators to manage timetables, view schedules, and perform user management tasks. It includes functionalities for adding new courses, managing instructor and room availability, generating timetables.

4. External Interface Requirements

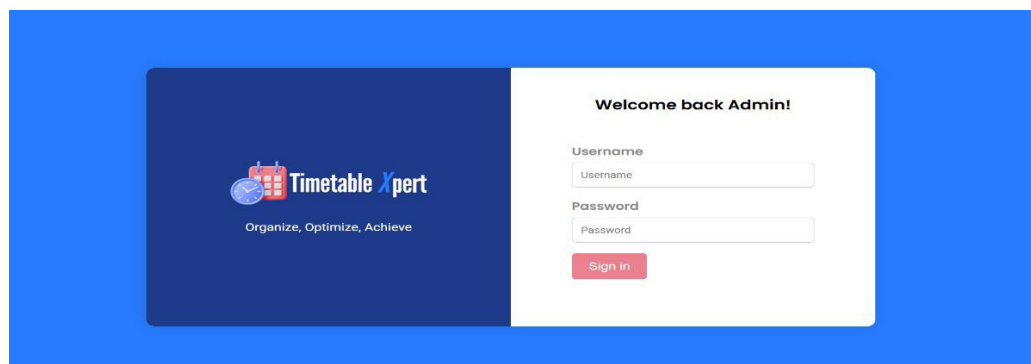
4.1 User Interfaces

The system will have a console-based interface suitable for interaction within a command-line environment on Windows.

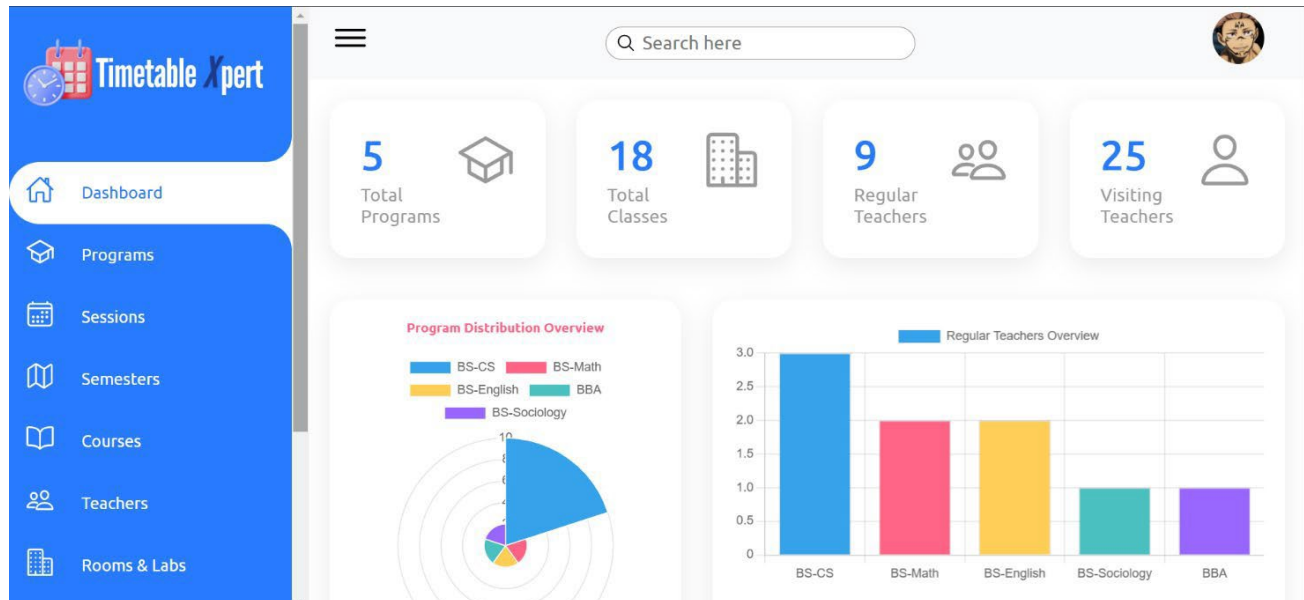
Landing Page



Login Page



Dashboard



Results

4.2.1 Department Wise

 **Institute of Arts and Sciences (Chiniot Campus) Govt College University Faisalabad**
Finalized Timetable Fall 2024
 Department: BS-CS Semester: BS-CS 4th Semester Dated: November 25, 2024

Time / Day	Monday	Tuesday	Wednesday	Thursday	Friday
08:00 - 09:00	Numerical Computing (CSI-412) \ 3(3-0) (Mr. Aqib) Room 4	Computer Organization and Assembly Language (CSI-406) \ 4(3-1) (Ms. Humaira) Room 4	Wireless Communication (CSI-410) \ 3(2-1) (Mr. Wasif Sharjeel) Room 4	Wireless Communication (CSI-410) \ 3(2-1) (Mr. Wasif Sharjeel) Room 4	Break
09:00 - 10:00	Wireless Communication (CSI-410) \ 3(2-1) (Mr. Wasif Sharjeel) Computer Lab	Data Structure and Algorithms (CSI-402) \ 4(3-1) (Mr. Saleem) Computer Lab	Data Structure and Algorithms (CSI-402) \ 4(3-1) (Mr. Saleem) Room 4	Numerical Computing (CSI-412) \ 3(3-0) (Mr. Aqib) Room 4	Database Systems (CCC-304) \ 4(3-1) (Mr. M Irfan) Room 4
10:00 - 11:00	Data Structure and Algorithms (CSI-402) \ 4(3-1) (Mr. Saleem) Room 4	Database Systems (CCC-304) \ 4(3-1) (Mr. M Irfan) Computer Lab	Computer Organization and Assembly Language (CSI-406) \ 4(3-1) (Ms. Humaira) Computer Lab	Computer Organization and Assembly Language (CSI-406) \ 4(3-1) (Ms. Humaira) Room 4	Computer Organization and Assembly Language (CSI-406) \ 4(3-1) (Ms. Humaira) Room 4
11:00 - 12:00	Break	Data Structure and Algorithms (CSI-402) \ 4(3-1) (Mr. Saleem) Room 4	Break	Database Systems (CCC-304) \ 4(3-1) (Mr. M Irfan) Room 4	Break
12:00 - 01:00	Break	Database Systems (CCC-304) \ 4(3-1) (Mr. M Irfan) Room 4	Break	Numerical Computing (CSI-412) \ 3(3-0) (Mr. Aqib) Room 4	Break

4.2.2 Teacher Wise

Institute of Arts and Sciences (Chiniot Campus) Govt College University Faisalabad



Finalized Timetable Fall 2024

Department: BS-CSName: Mr. Waqar Hussain - Time TableDated: December 01, 2024

Time / Day	Monday	Tuesday	Wednesday	Thursday	Friday
08:00 - 09:00	Break	BS-Math 5th Semester C++ Computer Lab	BBA 1st Semester Intro To Computing Application Computer Lab	BBA 1st Semester Intro To Computing Application Room 1	BS-Math 5th Semester C++ Room 1
09:00 - 10:00	BS-CS 3rd Semester Object Oriented Programming Room 3	BS-CS 3rd Semester Object Oriented Programming Room 3	BS-CS 3rd Semester Object Oriented Programming Computer Lab	Break	BS-Math 5th Semester C++ Room 1
10:00 - 11:00	BS-Math 5th Semester C++ Room 1	Break	Break	Break	Break
11:00 - 12:00	Break	Break	Break	Break	Break
12:00 - 01:00	Break	BS-CS 3rd Semester Object Oriented Programming Room 3	Break	Break	BBA 1st Semester Intro To Computing Application Room 1

4.2.3 Room Wise

Institute of Arts and Sciences (Chiniot Campus) Govt College University Faisalabad



Finalized Timetable Fall 2024

Department: BS-MathName: Room 1 - Time TableDated: November 25, 2024

Time / Day	Monday	Tuesday	Wednesday	Thursday	Friday
08:00 - 09:00	BS-Math 5th Semester Vector and Tensor Analysis (Ms. Noreen)	BS-Math 5th Semester C++ (Mr. Waqar Hussain)	BS-Math 5th Semester Vector and Tensor Analysis (Ms. Noreen)	BS-Math 5th Semester Algebra-I (Mr. Raheel)	BS-Math 5th Semester Algebra-I (Mr. Raheel)
09:00 - 10:00	BS-Math 5th Semester Algebra-I (Mr. Raheel)	BS-Math 5th Semester Complex Analysis (Dr. Mubashir)	BS-Math 5th Semester C++ (Mr. Waqar Hussain)	BS-Math 5th Semester Algebra-I (Mr. Raheel)	BS-Math 5th Semester Vector and Tensor Analysis (Ms. Noreen)
10:00 - 11:00	BS-Math 5th Semester Real Analysis I (Dr. Mubashir)	BS-Math 5th Semester C++ (Mr. Waqar Hussain)	BS-Math 5th Semester Complex Analysis (Dr. Mubashir)	BS-Math 5th Semester Vector and Tensor Analysis (Ms. Noreen)	Break
11:00 - 12:00	BS-Math 5th Semester Complex Analysis (Dr. Mubashir)	Break	BS-Math 5th Semester (Replica) C++ (Ms. Parveen)	BS-Math 5th Semester Real Analysis I (Dr. Mubashir)	BS-Math 5th Semester Complex Analysis (Dr. Mubashir)
12:00 - 01:00	BS-Math 5th Semester (Replica) C++ (Ms. Parveen)	BS-Math 5th Semester (Replica) Complex Analysis (Mr. Raheel)	Break	BS-Math 5th Semester Real Analysis I (Dr. Mubashir)	BS-Math 5th Semester Real Analysis I (Dr. Mubashir)

4.2.4 Lab Wise

Institute of Arts and Sciences (Chiniot Campus) Govt College University Faisalabad



Finalized Timetable Fall 2024

Department: BS-CS

Name: Computer Lab - Time Table

Dated: November 25, 2024

Time / Day	Monday	Tuesday	Wednesday	Thursday	Friday
08:00 - 09:00	Break	BS-CS 5th Semester Web Design and Development (Ms. Nosheen)	Break	Break	BS-CS 1st Semester Programming Fundamentals (Mr. Waseem Abbas)
09:00 - 10:00	BS-CS 4th Semester Wireless Communication (Mr. Wasif Sharjeel)	BS-CS 4th Semester Data Structure and Algorithms (Mr. Saleem)	BS-CS 2nd Semester Object Oriented Programming (Mr. Irshad Bhatti)	BS-CS 3rd Semester Computer Networks (Mr. Waseem Abbas)	BS-CS 3rd Semester Object Oriented Programming (Mr. Waqar Hussain)
10:00 - 11:00	BS-CS 5th Semester Operating System (Mr. Mustafa)	BS-CS 4th Semester Database Systems (Mr. M Irfan)	BS-CS 4th Semester Computer Organization and Assembly Language (Ms. Humaira)	BS-CS 2nd Semester Database Systems (Mr. Yaqoob)	Break
11:00 - 12:00	Break	Break	Break	BS-CS 1st Semester Introduction To Computing Applications (Ms. Rabia Kanwal)	Break
12:00 - 01:00	Break	Break	BS-CS 2nd Semester (Replica) Database Systems (Mr. Wasif Sharjeel)	Break	Break

4.2 Hardware Interfaces

The system will require a server with Java and SQL support. Users will access the system via a web browser on their devices.

4.3 Software Interfaces

The application will use **Java** for backend logic and **SQL** for database management. The frontend will be built using **HTML, CSS, and JavaScript**.

4.4 Communications Interfaces

The application will communicate with the SQL database for data storage and retrieval. It will also use external libraries for exporting timetables to Excel.

5. Other Non-Functional Requirements

5.1 Performance Requirements

The system should respond to user inputs promptly within the console environment. The application should efficiently process and display timetable information, ensuring a seamless user experience.

5.2 Safety Requirements

The system should ensure the security of user data stored in the database on the local machine.

5.3 Security Requirements

The application should implement measures to prevent unauthorized access to timetable data and other user information stored in the database.

5.4 Software Quality Attributes

Reliability: The system should function without errors, providing a stable experience for both administrators and students.

Usability: The console-based interface should be designed for ease of use, with clear prompts and instructions for users interacting via the command line.